



Cropland Coverage and Near Surface Conditions: A RegCM5 Study

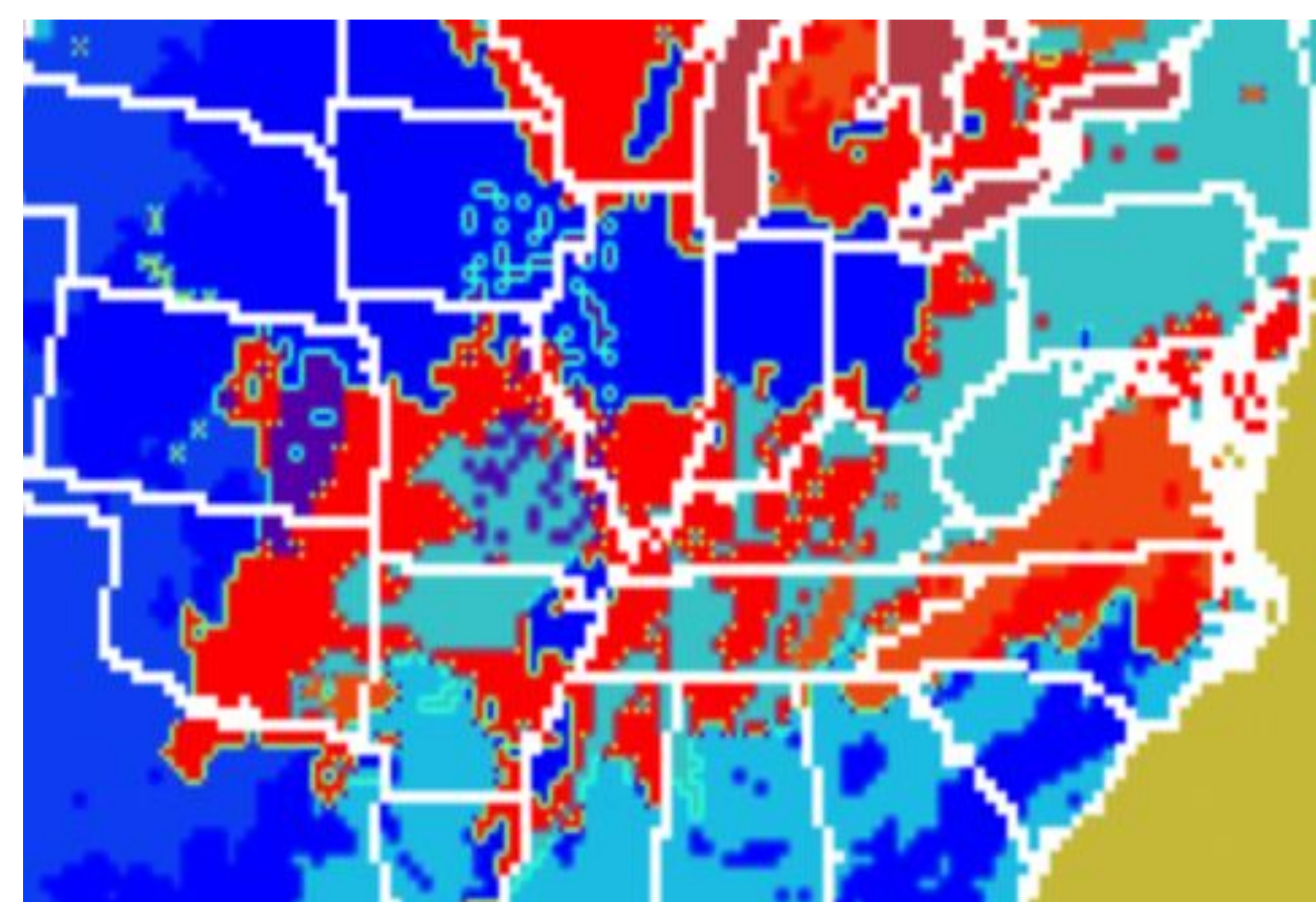
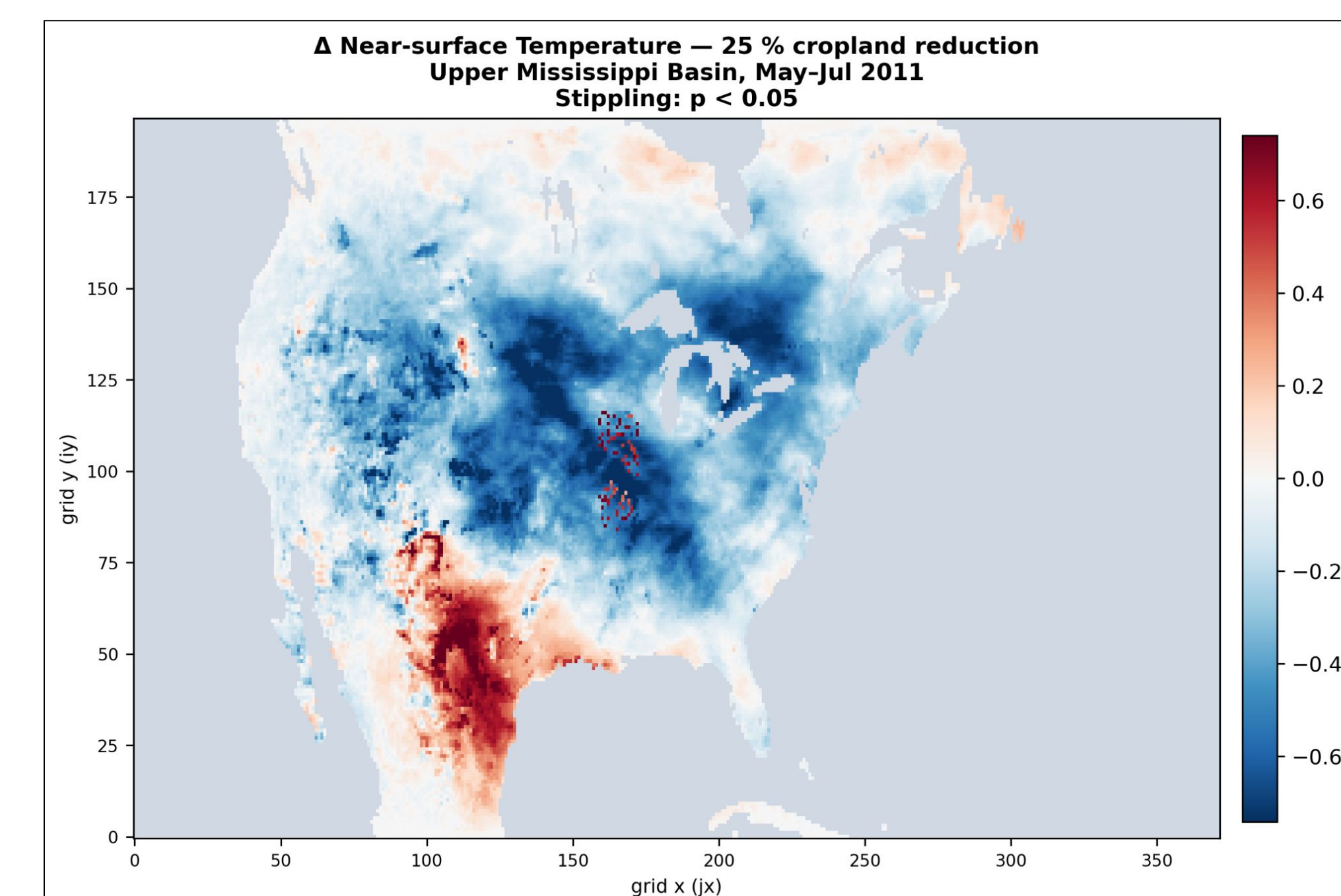
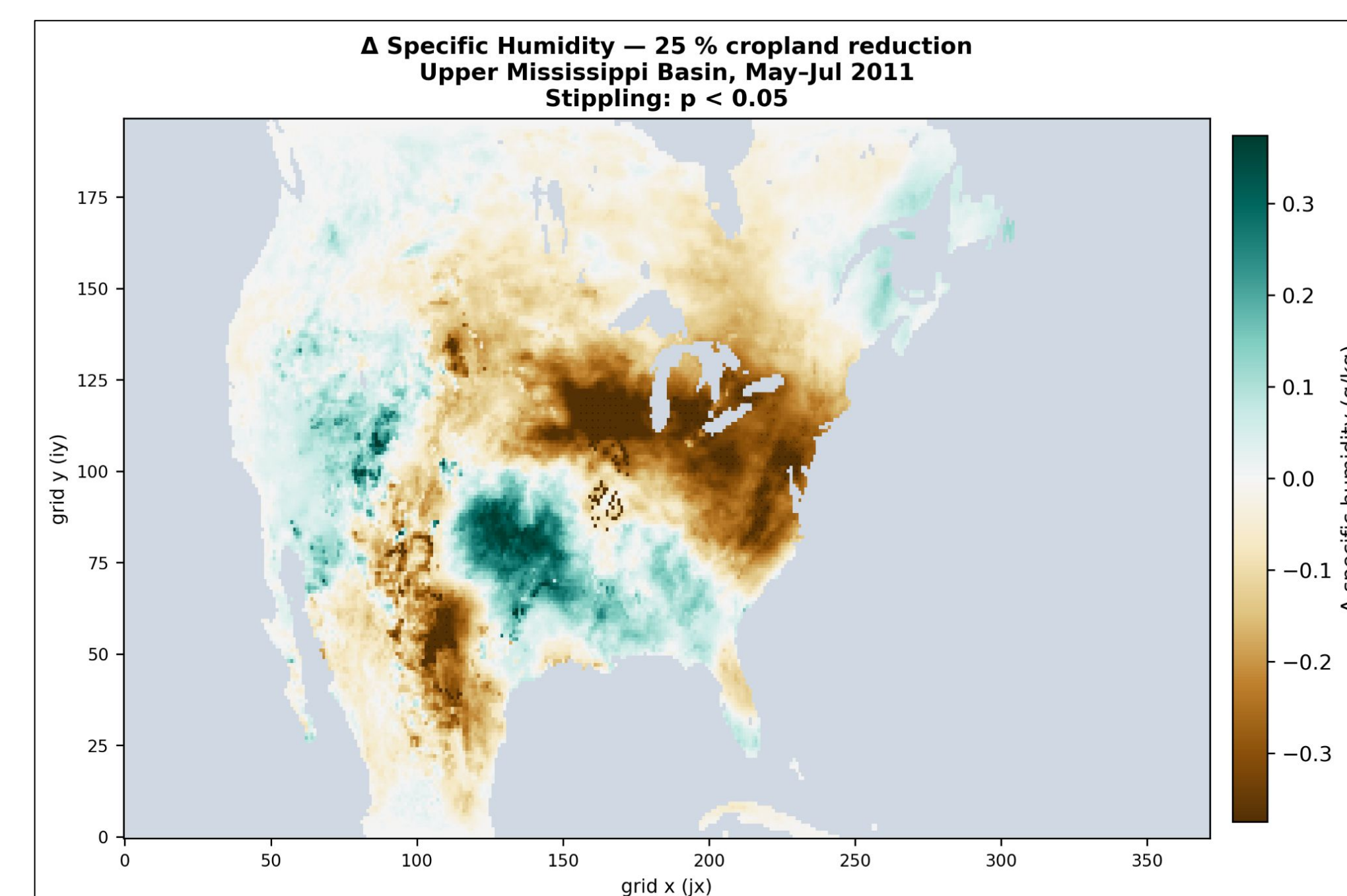
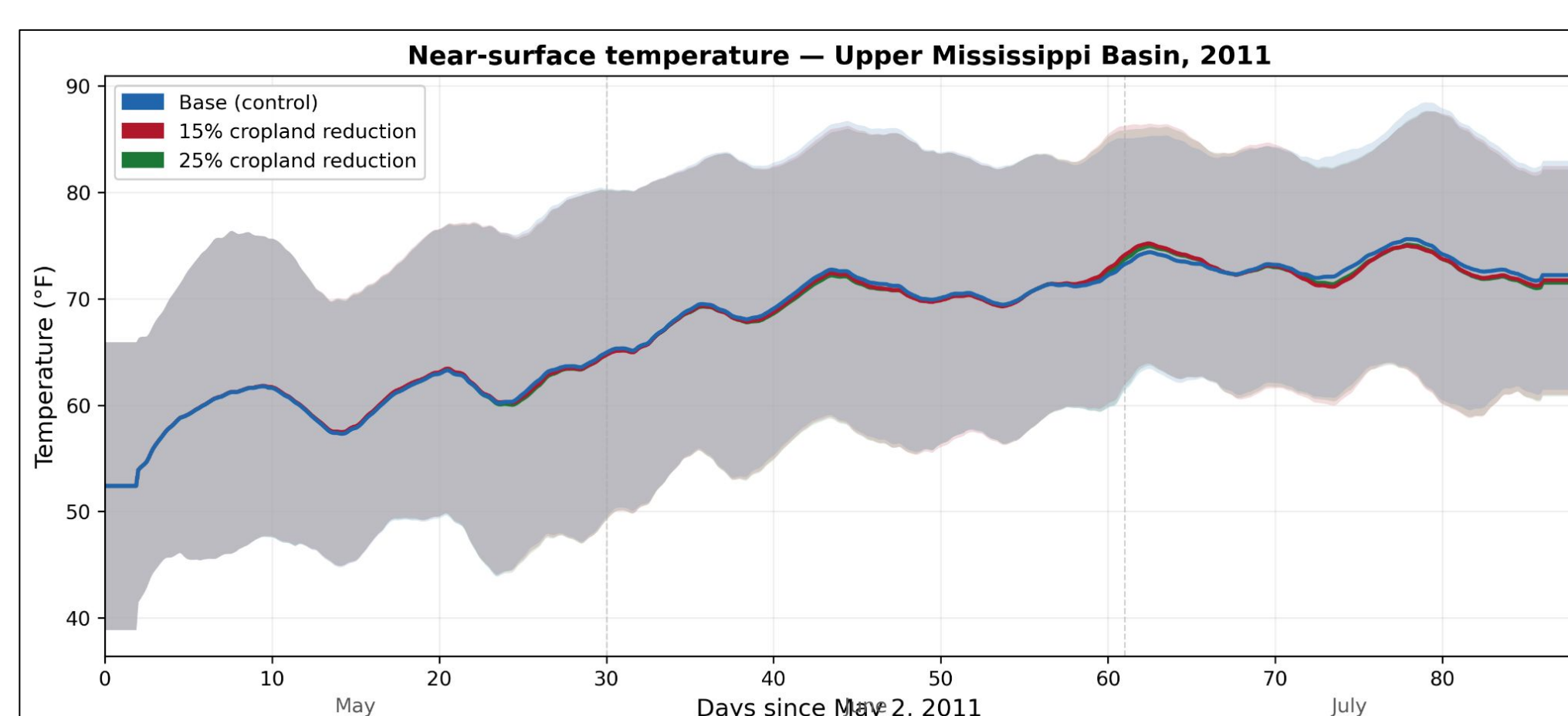
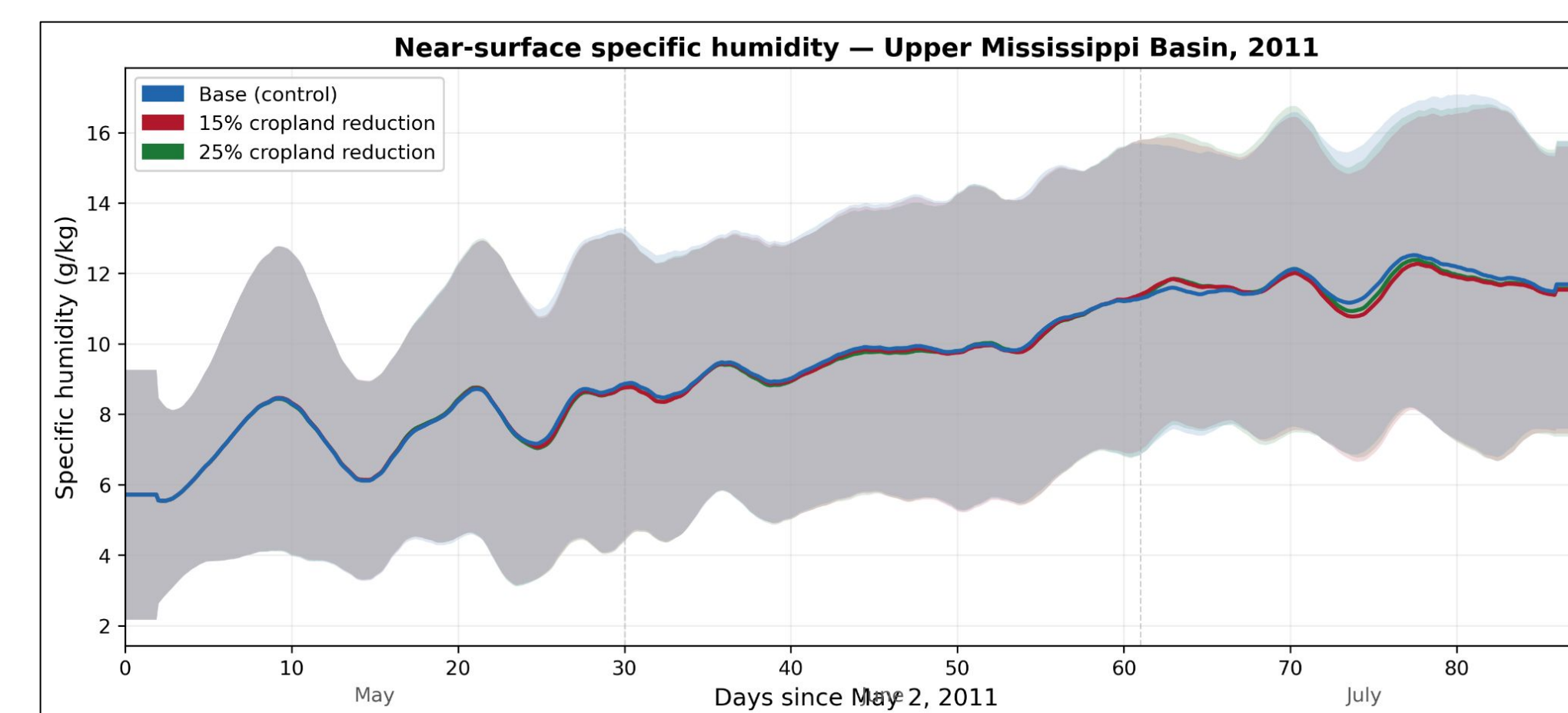
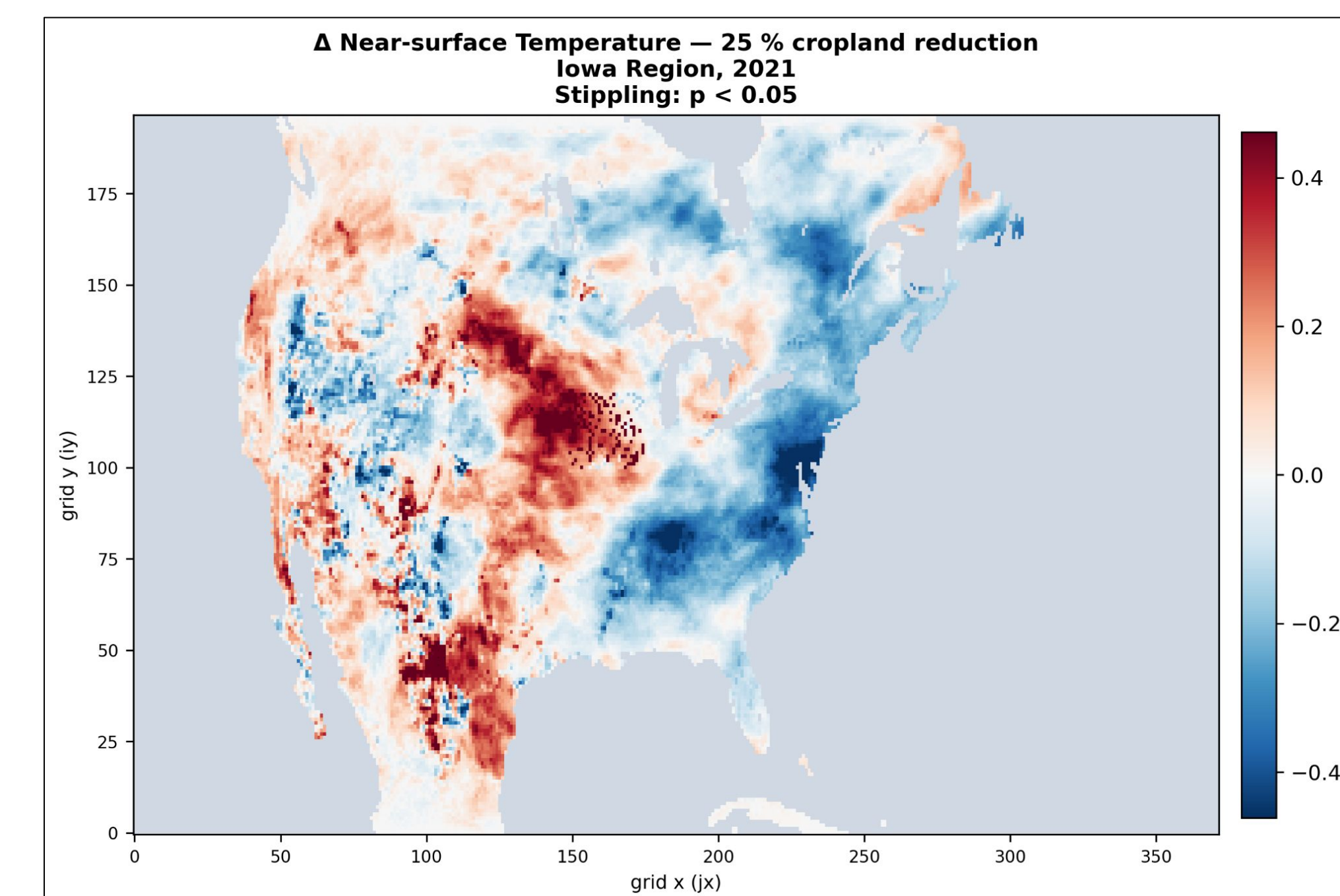
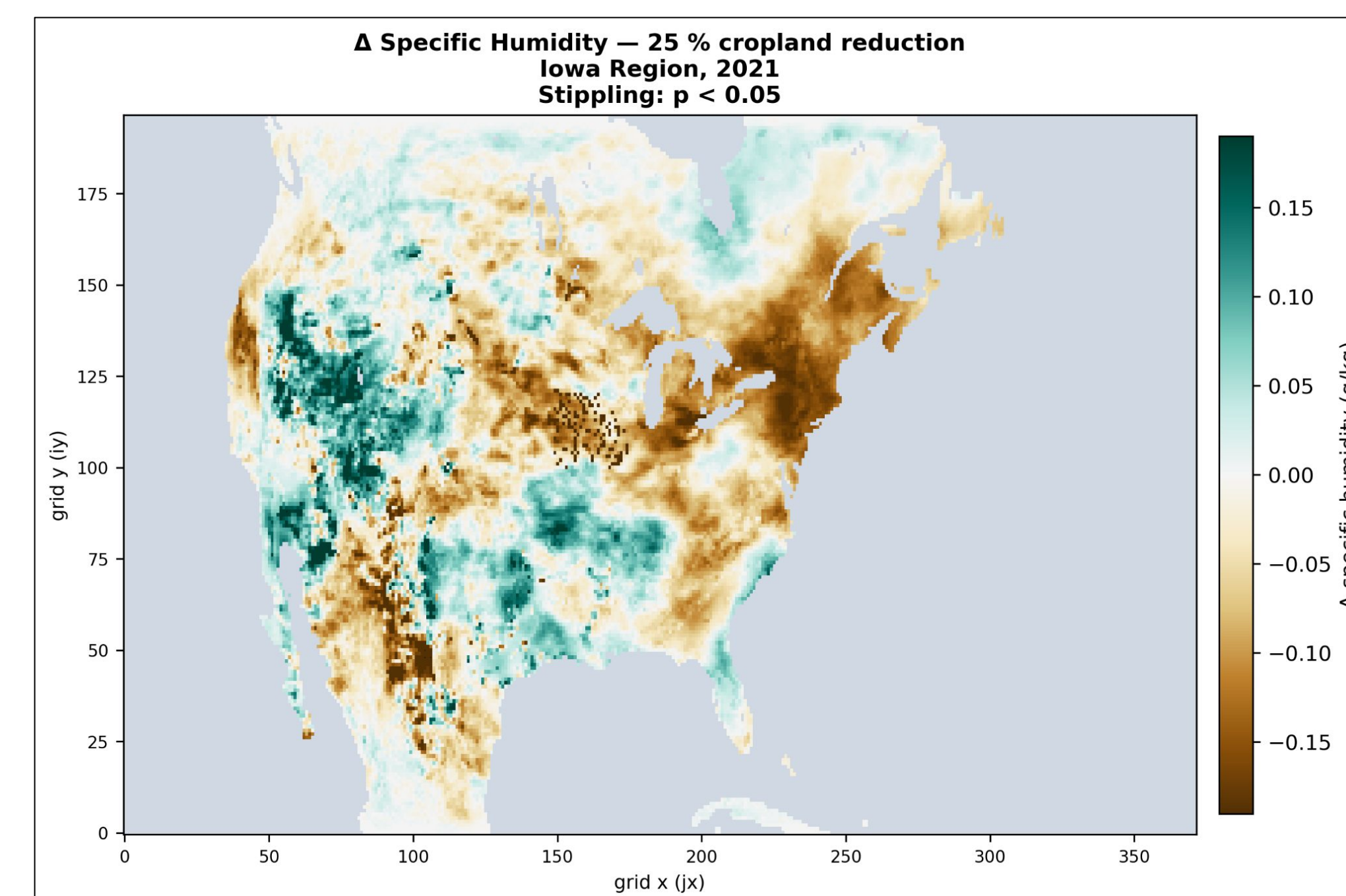
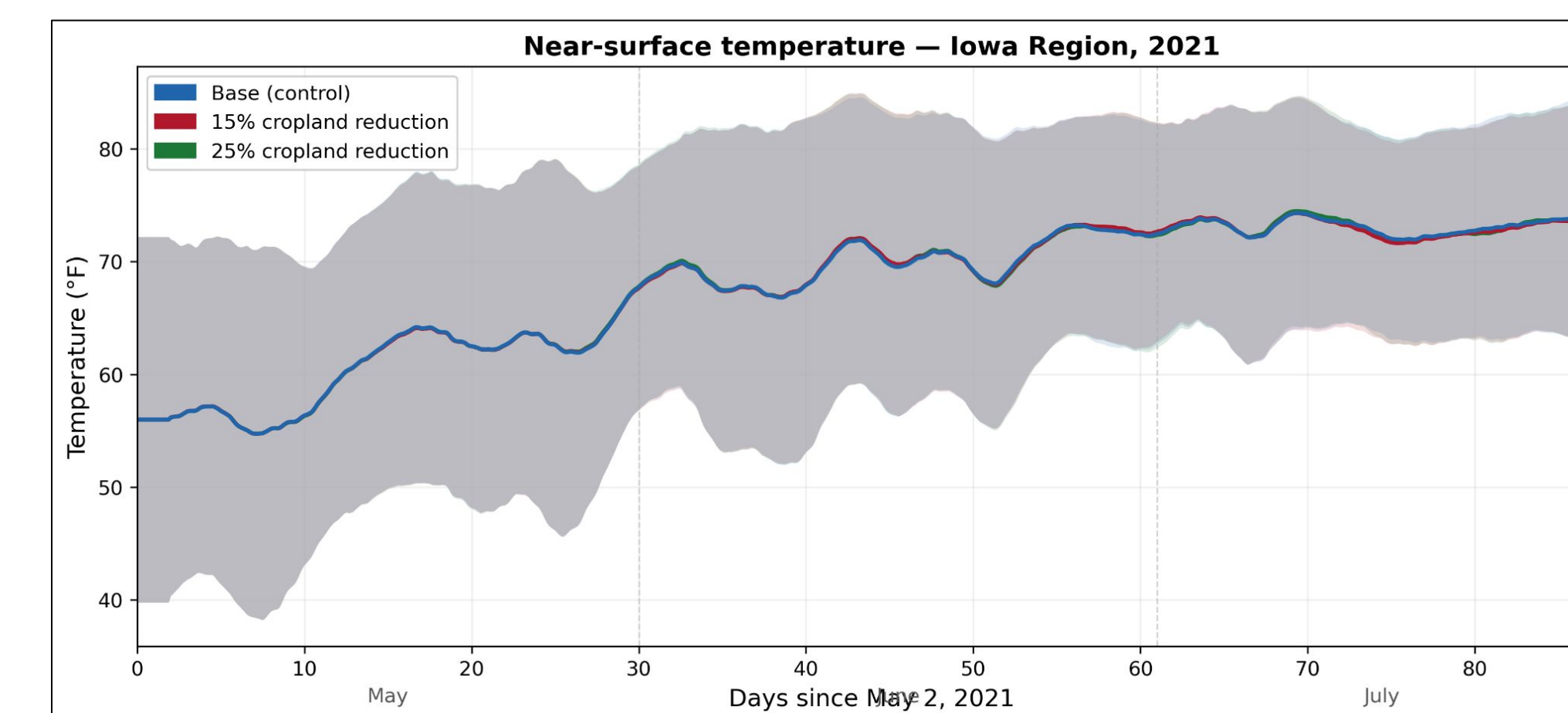
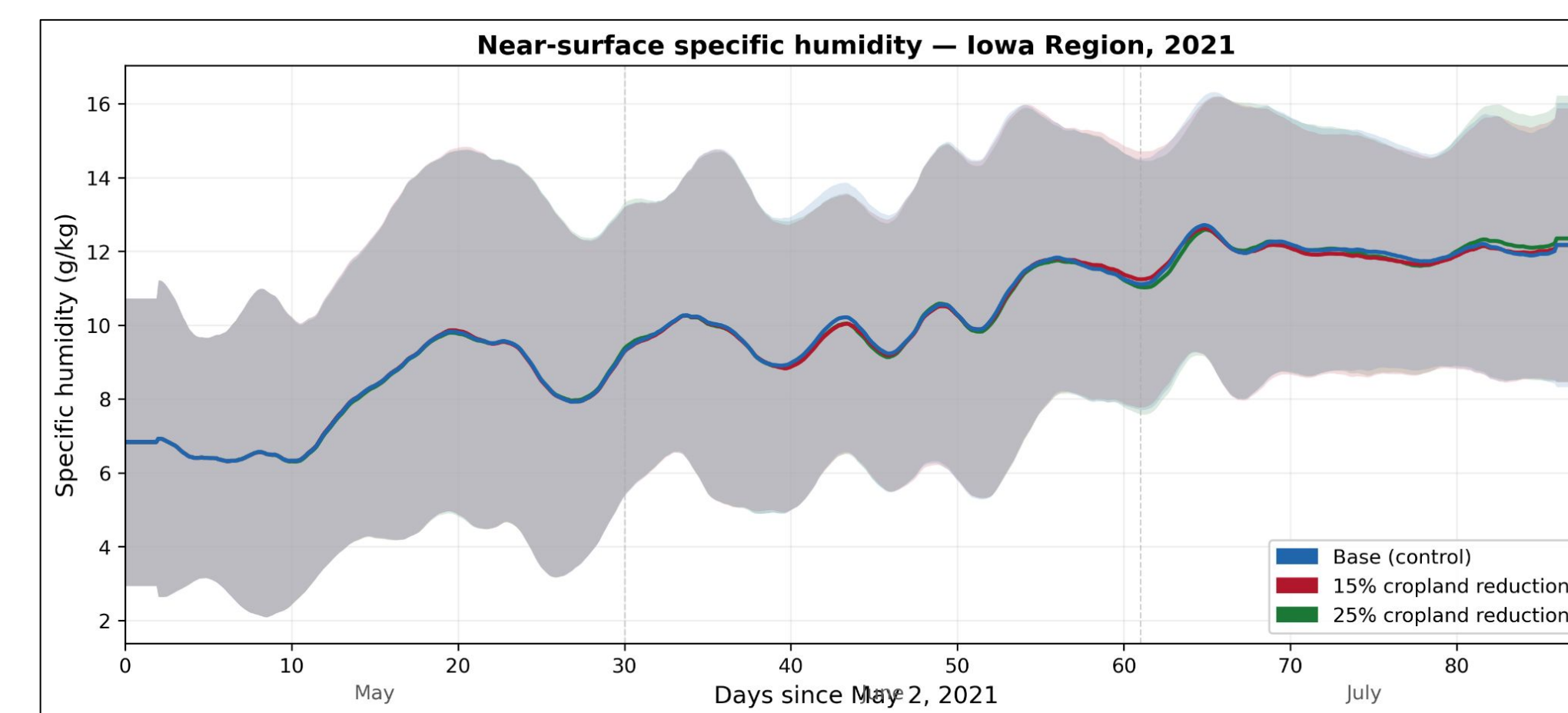
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Background

Regional climate systems are increasingly sensitive to land surface modifications, particularly in agriculturally intensive regions where land use decisions can significantly alter near-surface atmospheric conditions. This study aims to use the RegCM5 regional climate model to simulate the effects of varying cropland coverage on near-surface humidity across regional and national scales. By adjusting cropland coverage within the model, we plan to analyze how agricultural land use drives changes in surface moisture availability, as well as associated changes in heat index and regional weather patterns.

Approach

Simulations used the RegCM5 climate model to run a series of experiments in which cropland coverage was varied. Each experiment consists of eight ensemble members, generated by applying small random variations to the initial near-surface humidity to produce an ensemble spread. The resulting members are averaged to produce the test result. Simulations were performed during the summer in areas of: Iowa 2021, Chicago 1995, Upper Mississippi Basin 2011.



Results

Across all simulations, significant changes ($p < .05$) in near surface humidity and heat index emerge at both regional and national scales, though the magnitude of these differences varies considerably across experimentation.

Conclusion & Future Work

These results confirm that cropland coverage influences near-surface humidity and heat index at regional scales. Future work aims to extend this analysis to determine effects on crop yield, where temperature and moisture play a critical role^{1,3}.

Acknowledgements

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References

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